Worked Answer Key

Other sensible estimates and correctly explained mental strategies are also acceptable unless a method is specified.

Page 1. 1. $500 + 200 = 700$. 2. $8,000 - 3,000 = 5,000$. The hundreds digit in 7,842 is 8, so round up to 8,000; the hundreds digit in 3,196 is 1, so round down to 3,000.

Page 2. 1. $37 + 14 = 51$. 2. $10 \times 4 = 40$.

Page 3. 1. Estimate $400 \times 20 = 8,000$; exact $398 \times 21 = 7,960 + 398 = 8,358$. 2. Estimate $50 \times 20 = 1,000$ seats; exact $48 \times 19 = 960 - 48 = 912$ seats.

Page 4. 1. $200 + 300 + 400 = 900$. These nearby hundreds are easy to add mentally. 2. $4,800 \div 60 = 80$.

Page 5. 1. $(400 - 1) + 268 = 667$. 2. $2,500 + 1,806 = 4,306$; subtract the extra 3 added to 2,497: $4,306 - 3 = 4,303$.

Page 6. 1. $700 - 300 + 2 = 402$. 2. $2,796 + 204 = 3,000$, then $+2,000 = 5,000$; answer $2,204$.

Page 7. 1. Use $25 \times 4 = 100$ and $100 \times 4 = 400$, since $16 = 4 \times 4$. 2. $48 \times 5 = (48 \times 10) \div 2 = 240$. Multiplying by 5 is half of multiplying by 10.

Page 8. 1. $36 \div 9 = 4$, so $3,600 \div 9 = 400$. 2. Divide both numbers by 10 without changing the quotient: $2,400 \div 60 = 240 \div 6 = 40$.

Page 9. 1. $6,000 + 2,000 = 8,000$; low because the omitted parts, 482 and 759, add 1,241 to the exact sum. 2. $9,000 - 4,000 = 5,000$; low because the omitted parts contribute $876 - 321 = 555$ more to the exact difference, $5,555$.

Page 10. 1. $600 \times 7 = 4,200$, so $4,284$ is reasonable. 2. $5,000 \div 5 = 1,000$; 99 is not reasonable. (Exact answer is 990.)

Page 11. 1. Multiplication: estimate $6 \times 20 = 120$; exact $6 \times 24 = 144$. 2. Subtraction: estimate $240 - 100 = 140$; exact $238 - 97 = 141$.

Page 12. 1. Round each count and price to the nearest ten: $30 \times 20 + 10 \times 10 = 600 + 100 = 700$ dollars approximately; exact $27 \times 19 = 513$, $14 \times 12 = 168$, total 681 dollars. 2. Keep

8 crates exact; round 46 to 50 and 75 to 80: $8 \times 50 - 80 = 320$ apples approximately; exact $8 \times 46 = 368$, $368 - 75 = 293$ apples.

Page 13. 1. Estimate $200 + 200 = 400$; exact $186 + 217 = 403$. 2. Estimate $2,300 - 900 = 1,400$; exact $2,304 - 879 = 1,425$.

Page 14. 1. Lower: $32 \times 24 = 768$; upper: $32 \times 29 = 928$ students. 2. 550 through 649.

Page 15. 1. Total $4 \times 38 = 152$; $152 \div 8 = 19$ cans per box, remainder 0. Plan: multiply, then divide. 2. Plan: multiply to find the notebook cost, add the backpack cost, then subtract from the budget. Costs $3 \times 6 + 28 = 46$ dollars; $50 - 46 = 4$ dollars remain.

Page 16. 1. $20 + 20 + 30 + 30 = 100$ estimate; exact $18 + 24 + 31 + 27 = 100$. 2. $5,000 + 5,000 + 5,000 = 15,000$ (nearest thousand); exact $14,770$.

Page 17. 1. About $3 \times 15 = 45$ dollars; exact $14.95 \times 3 = 44.85$ dollars. 2. About $40 - 8 - 13 = 19$ dollars; exact $40.00 - 7.85 - 12.60 = 19.55$ dollars.

Page 18. 1. Compensation: move 1 from 1,001 to 999 to get $1,000 + 1,000 = 2,000$. Break apart: $999 + 1,000 + 1 = 1,999 + 1 = 2,000$. Sample comparison: compensation is more efficient for me because adding two thousands is one easy step. A justified preference for either method is acceptable. 2. Round both: $50 \times 20 = 1,000$. Round only 19: $51 \times 20 = 1,020$. Both are high; 1,000 is closer to the exact product, 969, since the estimates are 31 and 51 too high, respectively.

Page 19. 1. Estimate $9 \times 400 - 700 = 2,900$; exact $9 \times 418 = 3,762, 3,762 - 735 = 3,027$ occupied. The exact count is reasonable because it is close to the estimate of 2,900 occupied seats. 2. Estimate $300 + 400 + 300 + 200 + 400 = 1,600$; exact $286 + 419 + 305 + 198 + 392 = 1,600$ dollars. Yes, the total is $1,600 - 1,500 = 100$ dollars over the target. The exact total is reasonable because it matches the estimate.

Page 20. 1. Estimate $3,200 + 1,900 - 900 = 4,200$; exact $3,248 + 1,876 = 5,124$, $5,124 - 925 = 4,199$. 2. Total $24 \times 36 = 864$; $864 \div 9 = 96$ markers per classroom. Estimate $20 \times 40 \div 10 = 80$, so 96 is reasonable.